GENUS DOMINATES GRAPH GENUS FOR A MINIMAL NUMERICAL TYPE

THE STACKS PROJECT AUTHORS

Editorial note. Original author theorem and proof are retained. Only this target is counted; local supporting statements are not separate new entries.

Allowed standard prerequisites (editorial).

- Elementary graph homology, weighted symmetric intersection matrices and the explicit definitions of numerical type, minimality and genus retained in the source.. These are the named standard inputs; the selected main construction and its local nonroutine support remain in full.

Source reference. Original lemma 0C8S: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15377–15377. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0C8U: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15379–15379. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0C8T: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15378–15378. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

Source reference. Original lemma 0C5X: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15277–15277. Stable Stacks locator for declared standard input or surrounding context. No selected local proof is replaced by this pointer.

1. Numerical types

Part of the arguments will involve the combinatorics of the following data structures.

Definition 1.1. A *numerical type* T is given by

$$n, m_i, a_{ij}, w_i, g_i$$

where $n \geq 1$ is an integer and m_i, a_{ij}, w_i, g_i are integers for $1 \leq i, j \leq n$ subject to the following conditions

(1) $m_i > 0$, $w_i > 0$, $g_i \geq 0$,
(2) the matrix $A = (a_{ij})$ is symmetric and $a_{ij} \geq 0$ for $i \neq j$,
(3) there is no proper nonempty subset $I \subset \{1, \ldots, n\}$ such that $a_{ij} = 0$ for $i \in I$, $j \notin I$,
(4) for each i we have $\sum_j a_{ij} m_j = 0$, and
(5) $w_i | a_{ij}$.

Date: stacks-project:models.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

This is obviously a somewhat annoying type of structure to work with, but it is exactly what shows up in special fibres of proper regular models of smooth geometrically connected curves. Of course we only care about these types up to reordering the indices.

Definition 1.2. We say two numerical types n, m_i, a_{ij}, w_i, g_i and $n', m'_i, a'_{ij}, w'_i, g'_i$ are *equivalent types* if there exists a permutation σ of $\{1, \ldots, n\}$ such that $m_i = m'_{\sigma(i)}$, $a_{ij} = a'_{\sigma(i)\sigma(j)}$, $w_i = w'_{\sigma(i)}$, and $g_i = g'_{\sigma(i)}$.

A numerical type has a genus.

Lemma 1.3. *Let n, m_i, a_{ij}, w_i, g_i be a numerical type. Then the expression*

$$g = 1 + \sum m_i(w_i(g_i - 1) - \frac{1}{2}a_{ii})$$

is an integer.

Proof. To prove g is an integer we have to show that $\sum a_{ii}m_i$ is even. This we can see by computing modulo 2 as follows

$$\begin{aligned}
\sum\nolimits_i a_{ii}m_i &\equiv \sum\nolimits_{i,\ m_i \text{ odd}} a_{ii}m_i \\
&\equiv \sum\nolimits_{i,\ m_i \text{ odd}} \sum\nolimits_{j \neq i} a_{ij}m_j \\
&\equiv \sum\nolimits_{i,\ m_i \text{ odd}} \sum\nolimits_{j \neq i,\ m_j \text{ odd}} a_{ij}m_j \\
&\equiv \sum\nolimits_{i<j,\ m_i \text{ and } m_j \text{ odd}} a_{ij}(m_i + m_j) \\
&\equiv 0
\end{aligned}$$

where we have used that $a_{ij} = a_{ji}$ and that $\sum_j a_{ij}m_j = 0$ for all i. □

Definition 1.4. We say n, m_i, a_{ij}, w_i, g_i is a *numerical type of genus g* if $g = 1 + \sum m_i(w_i(g_i - 1) - \frac{1}{2}a_{ii})$ is the integer from Lemma 1.3.

We will prove below (Lemma 1.14) that the genus is almost always ≥ 0. But you can have numerical types with negative genus.

Lemma 1.5. *Let n, m_i, a_{ij}, w_i, g_i be a numerical type of genus g. If $n = 1$, then $a_{11} = 0$ and $g = 1 + m_1w_1(g_1 - 1)$. Moreover, we can classify all such numerical types as follows*

1. *If $g < 0$, then $g_1 = 0$ and there are finitely many possible numerical types of genus g with $n = 1$ corresponding to factorizations $m_1w_1 = 1 - g$.*
2. *If $g = 0$, then $m_1 = 1$, $w_1 = 1$, $g_1 = 0$ as in Lemma 0C8S.*
3. *If $g = 1$, then we conclude $g_1 = 1$ but m_1, w_1 can be arbitrary positive integers; this is case (0C8U) of Lemma 0C8T.*
4. *If $g > 1$, then $g_1 > 1$ and there are finitely many possible numerical types of genus g with $n = 1$ corresponding to factorizations $m_1w_1(g_1 - 1) = g - 1$.*

Proof. The lemma proves itself. □

Lemma 1.6. *Let n, m_i, a_{ij}, w_i, g_i be a numerical type of genus g. If $n > 1$, then $a_{ii} < 0$ for all i.*

Proof. Lemma 0C5X applies to the matrix A. □

Lemma 1.7. *Let n, m_i, a_{ij}, w_i, g_i be a numerical type of genus g. Assume $n > 1$. If i is such that the contribution $m_i(w_i(g_i - 1) - \frac{1}{2}a_{ii})$ to the genus g is < 0, then $g_i = 0$ and $a_{ii} = -w_i$.*

Proof. Follows immediately from Lemma 1.6 and $w_i > 0$, $g_i \geq 0$, and $w_i | a_{ii}$. □

Definition 1.8. Let n, m_i, a_{ij}, w_i, g_i be a numerical type. We say i is a (-1)-*index* if $g_i = 0$ and $a_{ii} = -w_i$.

We can "contract" (-1)-indices.

Lemma 1.9. *Let n, m_i, a_{ij}, w_i, g_i be a numerical type T. Assume n is a (-1)-index. Then there is a numerical type T' given by $n', m'_i, a'_{ij}, w'_i, g'_i$ with*

1. $n' = n - 1$,
2. $m'_i = m_i$,
3. $a'_{ij} = a_{ij} - a_{in}a_{jn}/a_{nn}$,
4. $w'_i = w_i/2$ *if* a_{in}/w_n *even and* a_{in}/w_i *odd and* $w'_i = w_i$ *else,*
5. $g'_i = \frac{w_i}{w'_i}(g_i - 1) + 1 + \frac{a_{in}^2 - w_n a_{in}}{2w'_i w_n}$.

Moreover, we have $g = g'$.

Proof. Observe that $n > 1$ for example by Lemma 1.5 and hence $n' \geq 1$. We check conditions (1) – (5) of Definition 1.1 for $n', m'_i, a'_{ij}, w'_i, g'_i$.

Condition (1) is immediate.

Condition (2). Symmetry of $A' = (a'_{ij})$ is immediate and since $a_{nn} < 0$ by Lemma 1.6 we see that $a'_{ij} \geq a_{ij} \geq 0$ if $i \neq j$.

Condition (3). Suppose that $I \subset \{1, \ldots, n-1\}$ such that $a'_{ii'} = 0$ for $i \in I$ and $i' \in \{1, \ldots, n-1\} \backslash I$. Then we see that for each $i \in I$ and $i' \in I'$ we have $a_{in}a_{i'n} = 0$. Thus either $a_{in} = 0$ for all $i \in I$ and $I \subset \{1, \ldots, n\}$ is a contradiction for property (3) for T, or $a_{i'n} = 0$ for all $i' \in \{1, \ldots, n-1\} \setminus I$ and $I \cup \{n\} \subset \{1, \ldots, n\}$ is a contradiction for property (3) of T. Hence (3) holds for T'.

Condition (4). We compute

$$\sum_{j=1}^{n-1} a'_{ij}m_j = \sum_{j=1}^{n-1}(a_{ij}m_j - \frac{a_{in}a_{jn}m_j}{a_{nn}}) = -a_{in}m_n - \frac{a_{in}}{a_{nn}}(-a_{nn}m_n) = 0$$

as desired.

Condition (5). We have to show that w'_i divides $a_{in}a_{jn}/a_{nn}$. This is clear because $a_{nn} = -w_n$ and $w_n | a_{jn}$ and $w_i | a_{in}$.

To show that $g = g'$ we first write

$$\begin{aligned} g &= 1 + \sum_{i=1}^{n} m_i(w_i(g_i - 1) - \frac{1}{2}a_{ii}) \\ &= 1 + \sum_{i=1}^{n-1} m_i(w_i(g_i - 1) - \frac{1}{2}a_{ii}) - \frac{1}{2}m_n w_n \\ &= 1 + \sum_{i=1}^{n-1} m_i(w_i(g_i - 1) - \frac{1}{2}a_{ii} - \frac{1}{2}a_{in}) \end{aligned}$$

Comparing with the expression for g' we see that it suffices if

$$w'_i(g'_i - 1) - \frac{1}{2}a'_{ii} = w_i(g_i - 1) - \frac{1}{2}a_{in} - \frac{1}{2}a_{ii}$$

for $i \leq n - 1$. In other words, we have

$$g'_i = \frac{2w_i(g_i - 1) - a_{in} - a_{ii} + a'_{ii} + 2w'_i}{2w'_i} = \frac{w_i}{w'_i}(g_i - 1) + 1 + \frac{a_{in}^2 - w_n a_{in}}{2w'_i w_n}$$

It is elementary to check that this is an integer ≥ 0 if we choose w'_i as in (4). □

Lemma 1.10. *Let n, m_i, a_{ij}, w_i, g_i be a numerical type. Let e be the number of pairs (i, j) with $i < j$ and $a_{ij} > 0$. Then the expression $g_{top} = 1 - n + e$ is ≥ 0.*

Proof. If not, then $e < n - 1$ which means there exists an i such that $a_{ij} = 0$ for all $j \neq i$. This contradicts assumption (3) of Definition 1.1. □

Definition 1.11. Let n, m_i, a_{ij}, w_i, g_i be a numerical type T. The *topological genus of* T is the nonnegative integer $g_{top} = 1 - n + e$ from Lemma 1.10.

We want to bound the genus by the topological genus. However, this will not always be the case, for example for numerical types with $n = 1$ as in Lemma 1.5. But it will be true for minimal numerical types which are defined as follows.

Definition 1.12. We say the numerical type n, m_i, a_{ij}, w_i, g_i of genus g is *minimal* if there does not exist an i with $g_i = 0$ and $a_{ii} = -w_i$, in other words, if there does not exist a (-1)-index.

We will prove that the genus g of a minimal type with $n > 1$ is greater than or equal to $\max(1, g_{top})$.

Lemma 1.13. *If n, m_i, a_{ij}, w_i, g_i is a minimal numerical type with $n > 1$, then $g \geq 1$.*

Proof. This is true because $g = 1 + \sum \Phi_i$ with $\Phi_i = m_i(w_i(g_i - 1) - \frac{1}{2}a_{ii})$ nonnegative by Lemma 1.7 and the definition of minimal types. □

Lemma 1.14. *If n, m_i, a_{ij}, w_i, g_i is a minimal numerical type with $n > 1$, then $g \geq g_{top}$.*

Proof. The reader who is only interested in the case of numerical types associated to proper regular models can skip this proof as we will reprove this in the geometric situation later. We can write

$$g_{top} = 1 - n + \frac{1}{2}\sum\nolimits_{a_{ij}>0} 1 = 1 + \sum\nolimits_i (-1 + \frac{1}{2}\sum\nolimits_{j \neq i,\ a_{ij}>0} 1)$$

On the other hand, we have

$$\begin{aligned} g &= 1 + \sum m_i(w_i(g_i - 1) - \frac{1}{2}a_{ii}) \\ &= 1 + \sum m_i w_i g_i - \sum m_i w_i + \frac{1}{2}\sum\nolimits_{i \neq j} a_{ij} m_j \\ &= 1 + \sum\nolimits_i m_i w_i (-1 + g_i + \frac{1}{2}\sum\nolimits_{j \neq i} \frac{a_{ij}}{w_i}) \end{aligned}$$

The first equality is the definition, the second equality uses that $\sum a_{ij} m_j = 0$, and the last equality uses that uses $a_{ij} = a_{ji}$ and switching order of summation. Comparing with the formula for g_{top} we conclude that the lemma holds if

$$\Psi_i = m_i w_i (-1 + g_i + \frac{1}{2}\sum\nolimits_{j \neq i} \frac{a_{ij}}{w_i}) - (-1 + \frac{1}{2}\sum\nolimits_{j \neq i,\ a_{ij}>0} 1)$$

is ≥ 0 for each i. However, this may not be the case. Let us analyze for which indices we can have $\Psi_i < 0$. First, observe that

$$(-1 + g_i + \frac{1}{2}\sum\nolimits_{j \neq i} \frac{a_{ij}}{w_i}) \geq (-1 + \frac{1}{2}\sum\nolimits_{j \neq i,\ a_{ij}>0} 1)$$

because a_{ij}/w_i is a nonnegative integer. Since $m_i w_i$ is a positive integer we conclude that $\Psi_i \geq 0$ as soon as either $m_i w_i = 1$ or the left hand side of the inequality is ≥ 0 which happens if $g_i > 0$, or $a_{ij} > 0$ for at least two indices j, or if there is a j with $a_{ij} > w_i$. Thus

$$P = \{i : \Psi_i < 0\}$$

is the set of indices i such that $m_i w_i > 1$, $g_i = 0$, $a_{ij} > 0$ for a unique j, and $a_{ij} = w_i$ for this j. Moreover

$$i \in P \Rightarrow \Psi_i = \frac{1}{2}(-m_i w_i + 1)$$

The strategy of proof is to show that given $i \in P$ we can borrow a bit from Ψ_j where j is the neighbour of i, i.e., $a_{ij} > 0$. However, this won't quite work because j may be an index with $\Psi_j = 0$.

Consider the set

$$Z = \{j : g_j = 0 \text{ and } j \text{ has exactly two neighbours } i, k \text{ with } a_{ij} = w_j = a_{jk}\}$$

For $j \in Z$ we have $\Psi_j = 0$. We will consider sequences $M = (i, j_1, \ldots, j_s)$ where $s \geq 0$, $i \in P$, $j_1, \ldots, j_s \in Z$, and $a_{ij_1} > 0, a_{j_1 j_2} > 0, \ldots, a_{j_{s-1} j_s} > 0$. If our numerical type consists of two indices which are in P or more generally if our numerical type consists of two indices which are in P and all other indices in Z, then $g_{top} = 0$ and we win by Lemma 1.13. We may and do discard these cases.

Let $M = (i, j_1, \ldots, j_s)$ be a maximal sequence and let k be the second neighbour of j_s. (If $s = 0$, then k is the unique neighbour of i.) By maximality $k \notin Z$ and by what we just said $k \notin P$. Observe that $w_i = a_{ij_1} = w_{j_1} = a_{j_1 j_2} = \ldots = w_{j_s} = a_{j_s k}$. Looking at the definition of a numerical type we see that

$$\begin{aligned} m_i a_{ii} + m_{j_1} w_i &= 0, \\ m_i w_i + m_{j_1} a_{j_1 j_1} + m_{j_2} w_i &= 0, \\ &\cdots\cdots \\ m_{j_{s-1}} w_i + m_{j_s} a_{j_s j_s} + m_k w_i &= 0 \end{aligned}$$

The first equality implies $m_{j_1} \geq 2m_i$ because the numerical type is minimal. Then the second equality implies $m_{j_2} \geq 3m_i$, and so on. In any case, we conclude that $m_k \geq 2m_i$ (including when $s = 0$).

Let k be an index such that we have a $t > 0$ and pairwise distinct maximal sequences $M_1, \ldots, M_t$ as above, with $M_b = (i_b, j_{b,1}, \ldots, j_{b,s_b})$ such that k is a neighbour of j_{b,s_b} for $b = 1, \ldots, t$. We will show that $\Phi_j + \sum_{b=1,\ldots,t} \Phi_{i_b} \geq 0$. This will finish the proof of the lemma by what we said above. Let M be the union of the indices occurring in M_b, $b = 1, \ldots, t$. We write

$$\Psi_k = -\sum\nolimits_{b=1,\ldots,t} \Psi_{i_b} + \Psi'_k$$

where

$$\begin{aligned} \Psi'_k = m_k w_k \left(-1 + g_k + \frac{1}{2} \sum\nolimits_{b=1,\ldots t} \left(\frac{a_{k j_{b,s_b}}}{w_k} - \frac{m_{i_b} w_{i_b}}{m_k w_k} \right) + \frac{1}{2} \sum\nolimits_{l \neq k,\ l \notin M} \frac{a_{kl}}{w_k} \right) \\ - \left(-1 + \frac{1}{2} \sum\nolimits_{l \neq k,\ l \notin M,\ a_{kl} > 0} 1 \right) \end{aligned}$$

Assume $\Psi'_k < 0$ to get a contradiction. If the set $\{l : l \neq k,\ l \notin M,\ a_{kl} > 0\}$ is empty, then $\{1, \ldots, n\} = M \cup \{k\}$ and $g_{top} = 0$ because $e = n - 1$ in this case and the result holds by Lemma 1.13. Thus we may assume there is at least one such l which contributes $(1/2)a_{kl}/w_k \geq 1/2$ to the sum inside the first brackets. For each $b = 1, \ldots, t$ we have

$$\frac{a_{k j_{b,s_b}}}{w_k} - \frac{m_{i_b} w_{i_b}}{m_k w_k} = \frac{w_{i_b}}{w_k} \left(1 - \frac{m_{i_b}}{m_k} \right)$$

This expression is $\geq \frac{1}{2}$ because $m_k \geq 2m_{i_b}$ by the previous paragraph and is ≥ 1 if $w_k < w_{i_b}$. It follows that $\Psi'_k < 0$ implies $g_k = 0$. If $t \geq 2$ or $t = 1$ and $w_k < w_{i_1}$, then $\Psi'_k \geq 0$ (here we use the existence of an l as shown above) which is a contradiction too. Thus $t = 1$ and $w_k = w_{i_1}$. If there at least two nonzero terms in the sum over l or if there is one such k and $a_{kl} > w_k$, then $\Psi'_k \geq 0$ as well. The final possibility is that $t = 1$ and there is one l with $a_{kl} = w_k$. This is disallowed as this would mean $k \in Z$ contradicting the maximality of M_1. $\square$